

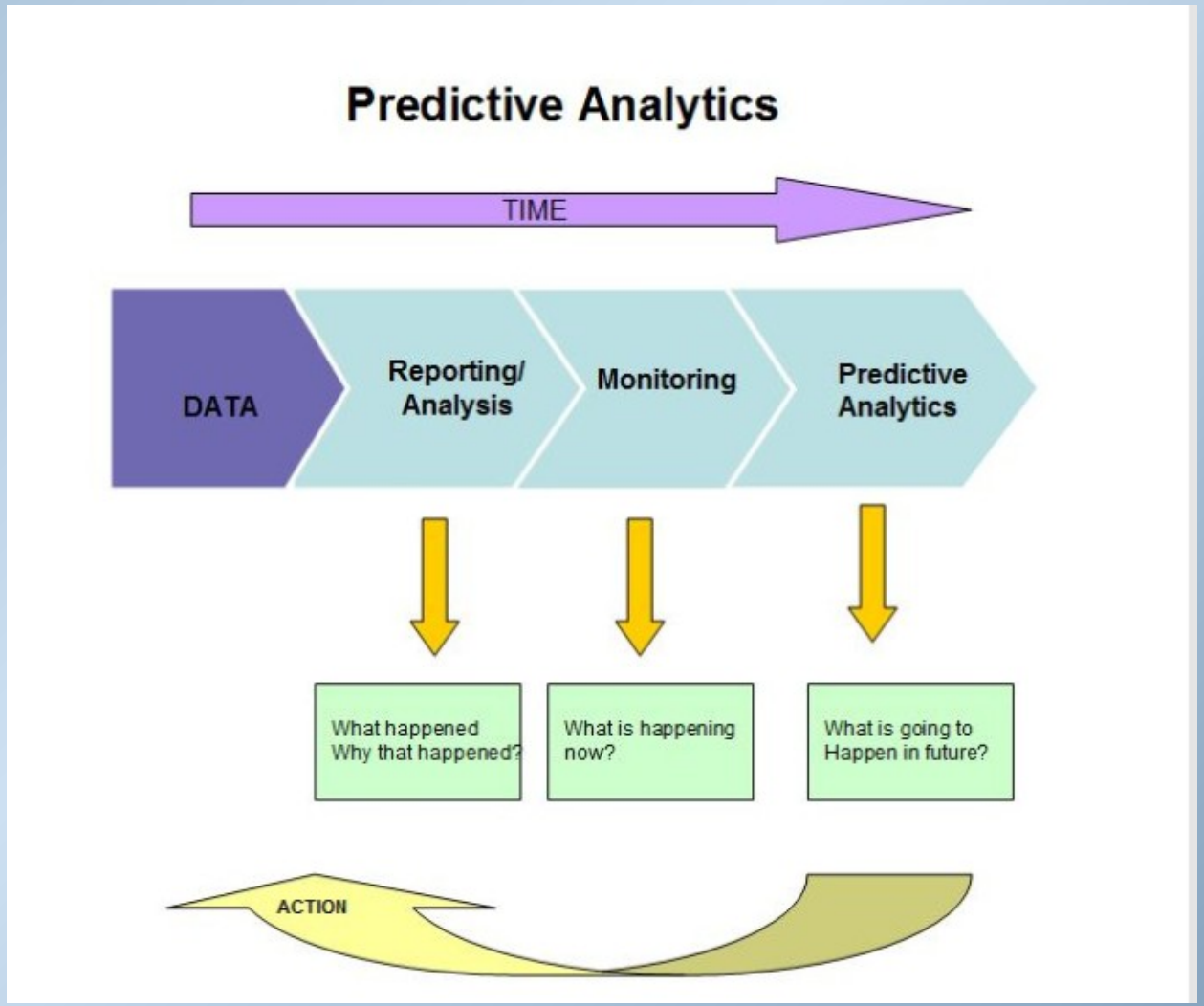
Predictive Analytics

Predictive Analytics-Introduction

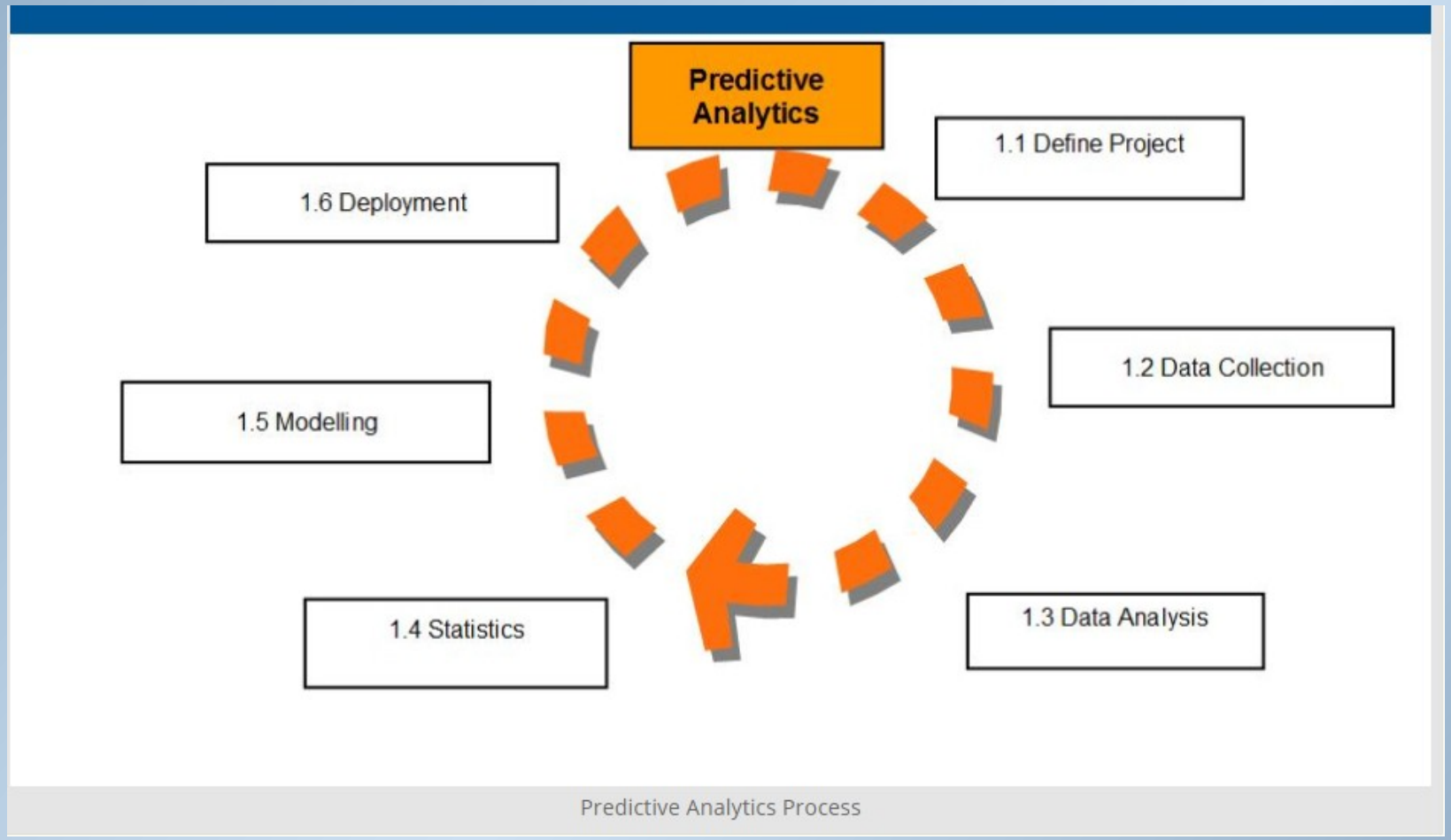
Predictive analytics is a form of advanced analytics that uses both new and historical data to forecast activity, behaviour and trends.

How it works?

- Uses many techniques from data mining, statistics, machine learning & artificial intelligence to analyze current data to make predictions about future.
- The patterns found in historical and transactional data can be used to identify risks and opportunities for future.



Predictive Analytics Process:



Predictive Analytics Process

Predictive Analytics Process: (Contd.)

- **Define Project:**

Define the project outcomes, deliverables, scoping of the effort, business objectives, identify the data sets which are going to be used.

- **Data Collection:**

Data mining for predictive analytics prepares data from multiple sources for analytics. This provides a complete view of customer interactions.

- **Data analysis:**

Data analysis is the process of inspecting ,cleaning, transforming, and modelling data with the objective of discoverable useful information, arriving at conclusions.

- **Statistics:**

Statistical Analysis enables to validate the assumptions, hypothesis and test them with using standard statistical models.

Predictive Analytics Process: (Contd.)

- **Modelling:**

Predictive modelling provides the ability to automatically create accurate predictive models about future.

- **Deployment:**

Predictive model deployment provides the option to deploy the analytical results in to the everyday decision making process to get results.

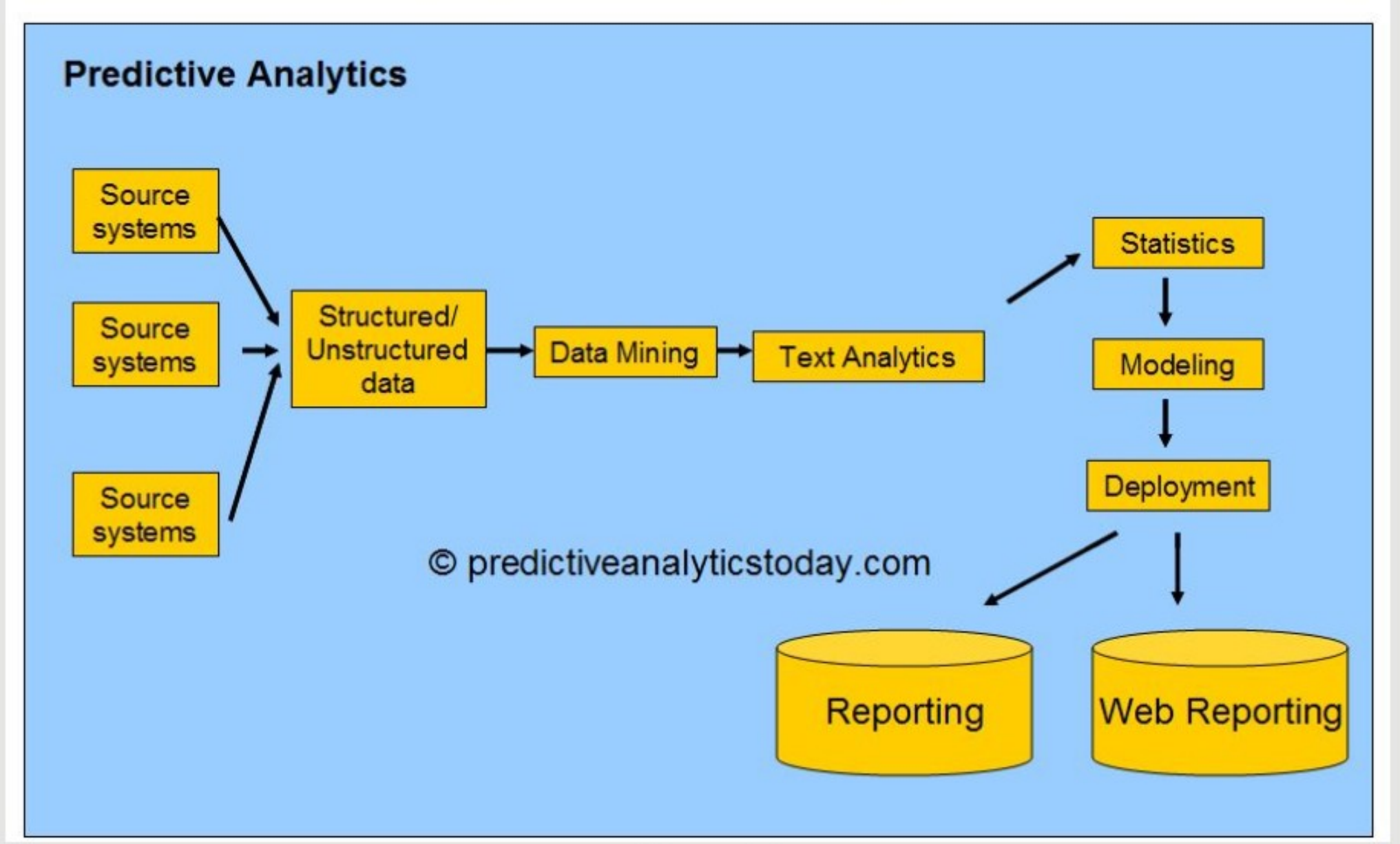
- **Model Monitoring:**

Models are monitored and managed to review the model performance to endure that it is providing the results expected.

Predictive Analytics: (Applications)

- Customer Relationship Management
- Health care
- Collection Analytics
- Cross sell
- Fraud detection
- Risk management
- Direct Marketing
- Underwriting

Predictive Analytics: (Applications)



Predictive analytics techniques that typically are applied:

- **Descriptive:** This technique summarizes what has happened in the past and attempts to analyse and characterize it.
- **Correlation.** Users can do correlation analysis to identify relationships and dependencies between different data variables to predict how they'll affect one another going forward.
- **Segmentation.** This technique is a way to analyze a large collection of entity data, such as a customer database, and organize it into smaller groups.

Predictive analytics techniques that typically are applied:

- **Regression.** This technique is designed to identify meaningful relationships among data variables, specifically looking at the connections between a dependent variable and other factors that may or may not affect it.
- **Association.** One more technique for highlighting relationships between data elements for predictive purposes is to look for ones that demonstrate affinity.
- **Classification.** Another means of separating different entities in a data set into related groups is to map them into predefined categories based on relevant characteristics or behaviours.

THANK YOU